Lecture 17 (15/08/21)

Diagonalization of a Matrix

$$Ax = \lambda x$$

* $\vec{x}$ is multiplied by a scalar, so all the difficulties of a matrix is swept away.

* If we want to multiply $\vec{x}$ by 100 times, we can actually form the eigenvector separately and its like having a diagonal matrix with no off diagonal intereconnection

In an $n \times n$ matrix, we have n separate eigenvalues and their corresponding eigenvectors $(x_1, x_2, \ldots, x_n)$

Representing all the eigenvalues in a single matrix,

$$S = \begin{bmatrix} \uparrow & \uparrow & \uparrow & & \uparrow \\ x_1 & x_2 & x_3 & \cdots & x_n \\ \downarrow & \downarrow & \downarrow & & \downarrow \end{bmatrix} \quad \text{(n dimensional space)}$$

Now,

$$AS = A\begin{bmatrix} \uparrow & \uparrow & \uparrow & & \uparrow \\ x_1 & x_2 & x_3 & \cdots & x_n \\ \downarrow & \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

$$= \begin{bmatrix} \uparrow & \uparrow & \uparrow & & \uparrow \\ \lambda_1 x_1 & \lambda_2 x_2 & \lambda_3 x_3 & \cdots & \lambda_n x_n \\ \downarrow & \downarrow & \downarrow & & \downarrow \end{bmatrix} \quad (Ax = \lambda x)$$

$$= \underbrace{\begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ x_1 & x_2 & \cdots & x_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}}_{S} \underbrace{\begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & \lambda_n \end{bmatrix}}_{\Lambda}$$

diagonal matrix

$$\therefore \boxed{AS = S\Lambda}$$

$$\Rightarrow ASS^{-1} = S\Lambda S^{-1}$$

$$\Rightarrow \boxed{A = S\Lambda S^{-1}}$$

S = Eigenvector matrix

S^{-1} = Inverse of S

Λ = diagonal matrix containing all eigenvalues

(S must be invertible)

To diagonalize a matrix,

we need,

Eigenvector matrix, S which must be invertible.

How to know if S is an invertible matrix?

In some cases the eigenvalues are the same, and for same eigenvalues we will get same eigenvectors,

If we have two identical eigenvectors (due to same eigenvalues), the will represent two columns of S, so it will be non-invertible (dependent column).

Why $A = S\Lambda S^{-1}$ form?

$$A^2 = A.A = S\Lambda \cancel{S^{-1}} . \cancel{S}\Lambda S^{-1} = S\Lambda^2 S^{-1}$$

$$A^3 = S\Lambda^3 S^{-1}$$

$$\therefore \boxed{A^k = S\Lambda^k S^{-1}}$$

$A = \begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix}$

$\lambda_1 = 1 \quad \lambda_2 = 6$ (triangular matrix)

For $\lambda = 1$,

$$\det(A - \lambda I) = \begin{vmatrix} 1-1 & 5 \\ 0 & 6-1 \end{vmatrix} = \begin{vmatrix} 0 & 5 \\ 0 & 5 \end{vmatrix}$$

$$\therefore x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

For $\lambda = 6$,

$$\det(A - \lambda I) = \begin{vmatrix} 1-6 & 5 \\ 0 & 6-6 \end{vmatrix} = \begin{vmatrix} -5 & 5 \\ 0 & 0 \end{vmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \therefore S^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$\therefore A^k = \overbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}^{S} \overbrace{\begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}^k}^{\Lambda} \overbrace{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}}^{S^{-1}}$$

$$= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 6^k \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Use of this kind of equation

$$u_{k+1} = A\,u_k \qquad \text{(sequence)}$$

$$\underline{u_k} = A^k \underline{u_0}$$

Fibonacci Sequence

0, 1, 1, 2, 3, 5, 8 ...

100th term ?

$$F_{k+2} = F_{k+1} + F_k$$

represent as $u_k = A^k u_0$

$$\begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} F_{k+1} + F_k \\ F_{k+1} \end{bmatrix}$$

$$u_{k+1} = \begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} F_{k+1} + F_k \\ F_{k+1} \end{bmatrix}}$$

$$= \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}}_{u_k}$$

$$\therefore \quad U_{k+1} = A \cdot U_k$$

$$\begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

$$\rightarrow \quad U_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{matrix} \rightarrow F_1 \\ \rightarrow F_0 \end{matrix}$$

$$U_1 = A \cdot U_0 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{matrix} \rightarrow F_2 \\ \rightarrow F_1 \end{matrix}$$

$$\therefore \quad U_2 = A^2 U_0 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{matrix} \rightarrow F_3 \\ \rightarrow F_2 \end{matrix}$$

$$U_{100} = \begin{bmatrix} F_{101} \\ F_{100} \end{bmatrix} = A^{100} U_0$$

Now, $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = -\lambda + \lambda^2 - 1 = \lambda^2 - \lambda - 1 = 0$$

$$\therefore \lambda_1 = \frac{1+\sqrt{5}}{2} \qquad \lambda_2 = \frac{1-\sqrt{5}}{2}$$

Now, $a_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} \qquad a_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}$

$$\therefore S = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}$$

$$S^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \qquad \Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$\therefore A = S \Lambda S^{-1}$$

$$A = S \Lambda S^{-1}$$

$$= \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$$

$$= \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$$

$$\therefore A^k = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda^k \end{bmatrix} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$$

$$U_k = A^k u_0$$

$$\begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = A^k \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k \\ -\lambda_2^k \end{bmatrix}$$

$$\begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1^{k+1} - \lambda_2^{k+1} \\ \lambda_1^k - \lambda_2^k \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{K+1} \\ F_K \end{bmatrix} = \begin{bmatrix} \dfrac{\lambda_1^{K+1} - \lambda_2^{K+1}}{\lambda_1 - \lambda_2} \\ \dfrac{\lambda_1^{K} - \lambda_2^{K}}{\lambda_1 - \lambda_2} \end{bmatrix}$$

$$\therefore \; F_K = \frac{\lambda_1^{K} - \lambda_2^{K}}{\lambda_1 - \lambda_2}$$

$K = 100,$

$$F_{100} = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^{100} - \left(\frac{1-\sqrt{5}}{2}\right)^{100}}{\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}}$$

$$= \frac{(1+\sqrt{5})^{100} - (1-\sqrt{5})^{100}}{2^{100} \cdot (\sqrt{5})}$$

Homework

$$u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

$$f_{k+2} = f_{k+1} + 2 f_k$$

f_k ? $\quad k = 1005 \quad$ Using Diagonalization.

Solution

$$f_{k+2} = f_{k+1} + 2 f_k$$

$$f_{k+1} = f_{k+1}$$

$$\therefore \begin{bmatrix} f_{k+2} \\ f_{k+1} \end{bmatrix} = \begin{bmatrix} f_{k+1} + 2 f_k \\ f_{k+1} \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} f_{k+2} \\ f_{k+1} \end{bmatrix}}_{u_{k+1}} = \underbrace{\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} f_{k+1} \\ f_k \end{bmatrix}}_{u_k}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 1 & -\lambda \end{bmatrix}$$

$$\therefore \det(A - \lambda I) = \lambda^2 - \lambda - 2 = 0$$

$$\therefore \lambda_1 = 2 \qquad \lambda_1 = -1$$

$$\therefore x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\therefore S = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \quad S^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{bmatrix} \quad \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A = \underbrace{\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}}_{S} \cdot \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}}_{\Lambda} \frac{1}{3} \underbrace{\begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}}_{S^{-1}}$$

$$= \frac{1}{3} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}$$

$$u_k = A^k \cdot u_0$$

$$= S \Lambda^k S^{-1} u_0$$

$$= \frac{1}{3}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2^k & 0 \\ 0 & (-1)^k \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2^k & 0 \\ 0 & (-1)^k \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2^k \\ (-1)^{k+1} \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 2^{k+1} + (-1)^{k+2} \\ 2^k + (-1)^{k+1} \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{2^{k+1} + (-1)^{k+2}}{3} \\ \dfrac{2^k + (-1)^{k+1}}{3} \end{bmatrix} \begin{matrix} \rightarrow f_{k+1} \\ \rightarrow F_k \end{matrix}$$

$$\therefore f_k = \frac{2^k + (-1)^{k+1}}{3}$$

$$\therefore f_{1005} = \frac{2^{1005} + (-1)^{1005+1}}{3} = \frac{2^{1005} + 1}{3} \text{ (Ans.)}$$